

AI-GENERATED MODEL PAPER

Subject: Database Management Systems

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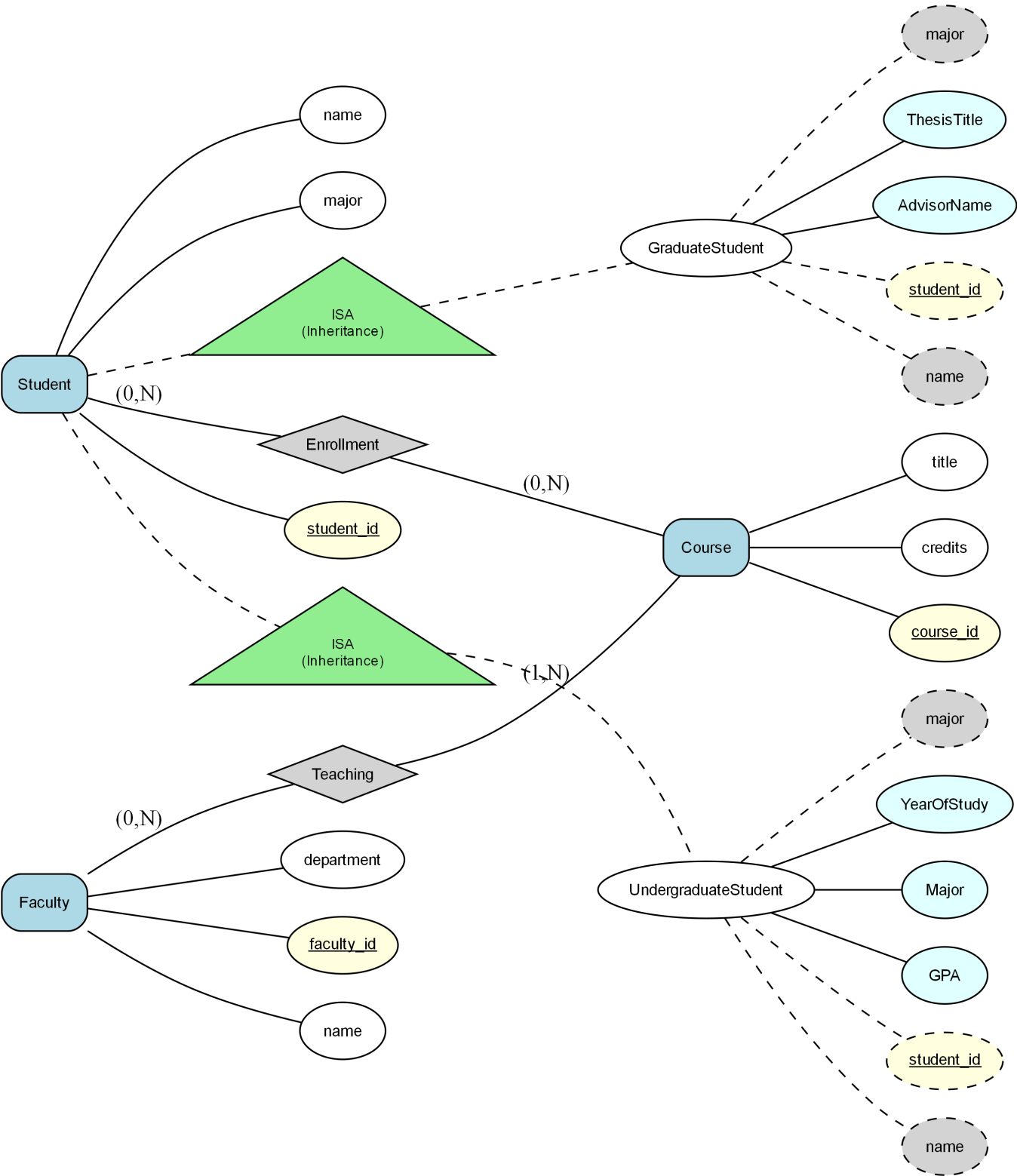
Total Marks: 100

Examination Paper

Question Q1 (20 marks)

Consider the following Enhanced Entity-Relationship (EER) diagram:

Figure: EER



- a) Convert the following EER model into the relational model. Indicate the primary keys and the foreign keys of the resulting relations clearly. (17)
- b) What is the best option for mapping the ISA hierarchies in the diagram? Justify your answer. (3)

Question Q2 (15 marks)

Consider a relation $R(\text{EmployeeID}, \text{EmployeeName}, \text{DepartmentID}, \text{DepartmentName}, \text{Salary})$ with the following set of functional dependencies F over R : $F = \{\{ \text{EmployeeID} \rightarrow \text{EmployeeName}, \text{DepartmentID} \rightarrow \text{DepartmentName}, \text{EmployeeID} \rightarrow \text{Salary} \}\}$.

- a) Draw the functional dependency diagram. (1)
- b) Use the attribute closure and identify the primary key for the given relation. (1)
- c) If we insert a row $\langle 101, \text{John Doe}, 4, \text{HR}, 75000 \rangle$ into the given relation, what is the update anomaly that is violated? Give reasons. (2)
- d) Assume that the relation has a row, $\langle 102, \text{Jane Smith}, 3, \text{IT}, 65000 \rangle$. If we remove the record for Jane Smith, what is the possible update anomaly that can be violated? Give reasons. (2)
- e) Using the above functional dependencies, design a set of 3NF relations for the above given relation. Show clearly each stage in deriving the 3NF relations. Also identify the normal form at each of these stages. (3)
- f) Do the update anomalies of part (c) and (d) hold in the 3NF relations? Discuss briefly. (2)
- g) Are the 3NF relations derived in part (e) still in 3NF? Justify your answer. (2)
- h) Are the 3NF relations derived in part (e) in BCNF? If not, bring the relations to BCNF. (2)

Question Q3 (25 marks)

A university is managing its course and student enrollment data using a relational database system. The database contains several relations, including $\text{Courses}(\text{course_id: string}, \text{course_name: string}, \text{credits: integer})$ and $\text{Students}(\text{student_id: string}, \text{student_name: string}, \text{major: string})$. In addition, the university requires constraints to ensure data integrity for managing course enrollments in a relation called $\text{Enrollments}(\text{course_id: string}, \text{student_id: string}, \text{semester: string})$. Enrollments must refer to valid course_id and student_id from the respective relations.

- a) Accept or refute the statement about using relational schemas to define the structure

- of a database. Justify your answer. (2)
- b) Which type of statements is used in JDBC API to execute updates on the database? Briefly explain when this type of statement will be used. (2)
- c) Briefly explain the methods available for transferring data between the Courses and Students tables. (2)
- d) Briefly explain how data integrity can be ensured for the Enrollments relation in the university database. (3)
- e) A new requirement states that each course must have a unique identifier. Write a T-SQL statement to create the Courses table including this requirement. (2)
- f) Provide permissions to a database user to manage the Courses and Enrollments relations. Write a T-SQL statement to grant this access. (5)
- g) Assuming the user is 'student101', write T-SQL statements to allow this user to retrieve data from the Courses table. (3)
- h) Assuming the user 'admin' needs to manage data integrity, write T-SQL statements to enforce foreign key relationships in the Enrollments table. (3)
- i) Assuming the user has been given access to the database, write T-SQL statements to allow him to insert new student enrollment records in the Enrollments table. (3)

Question Q4 (40 marks)

In a university database, there are three main entities: Student, Course, and Instructor. Each Student can enroll in multiple Courses, and each Course is taught by a single Instructor. The Student entity has attributes like StudentID, Name, and Major. The Course entity includes CourseID, Title, and Credits, while the Instructor entity contains InstructorID, Name, and Department. The enrollment relationship has an attribute for Grade, representing the Student's performance in a Course.

- a) Identify all the entities and their attributes in the given scenario. (10)
- b) Identify the relationships and their cardinalities among the entities in the scenario. (10)
- c) Draw an ER diagram to represent the entities, attributes, and relationships identified in the scenario. (10)
- d) Map the identified entities and relationships to a relational schema, indicating primary keys. (10)